# US Patent & Trademark Office
# Patent Public Search | Text View


| | |
|---|---|
| United States Patent | 12413329 |
| Kind Code | B2 |
| Date of Patent | September 09, 2025 |
| Inventor(s) | Ben-Ezra; Yosef et al. |


## System for pulsed laser optical data transmission with receiver recoverable clock


### Abstract

A system, comprising an optical transmitter configured to modulate a pulsed laser signal based on a plurality of data signals to form a plurality of modulated optical data signals, optical time division multiplex the plurality of modulated optical data signals to obtain a multiplexed optical data signal, and transmit the multiplexed optical data signal together with an optical clock signal derived from the pulsed laser signal; and an optical receiver that includes a clock recovery branch configured to recover the optical clock signal into a plurality of phase-shifted recovered optical clock signals and a data branch with a plurality of saturable absorbers (SA), each SA being in optical communication with one of the phase-shifted recovered optical clock signals and configured to receive the multiplexed optical data signal and to transmit of one of the plurality of modulated optical data signals according to the recovered optical clock signal.



**Inventors:** **Ben-Ezra; Yosef (Petah Tikva, IL), Ben-Haim; Yaniv (Petah Tikva, IL)**

**Applicant:** **NewPhotonics Ltd.** (Petah Tikva, IL)



**Family ID:** **82654207**

**Assignee:** **NewPhotonics Ltd. (Petah Tikva, IL)**

**Appl. No.:** **18/675378**

**Filed:** **May 28, 2024**

### Prior Publication Data

| Document Identifier | Publication Date |
|---|---|
| US 20240313879 A1 | Sep. 19, 2024 |

### Related U.S. Application Data

continuation parent-doc US 17916807 US 12034527 WO PCT/IB2022/050063 20220105 child-doc US 18675378
us-provisional-application US 63142684 20210128

---

## Publication Classification

**Int. Cl.:** **H04J14/08** (20060101); **H04B10/50** (20130101); **H04B10/524** (20130101); **H04B10/60** (20130101); H04B10/40 (20130101)

**U.S. Cl.:**

CPC **H04J14/08** (20130101); **H04B10/524** (20130101); **H04B10/60** (20130101); H04B10/40 (20130101); H04B10/50 (20130101)

## Field of Classification Search

**USPC:** None

---

## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 5519526 | 12/1995 | Chua | 398/139 | H04J 14/005 |
| 6118566 | 12/1999 | Price | 398/1 | H04J 14/0298 |
| 6411415 | 12/2001 | Smets et al. | N/A | N/A |
| 6483620 | 12/2001 | Epworth | 398/154 | H04B 10/2569 |
| 2003/0058500 | 12/2002 | Sugawara | N/A | N/A |
| 2004/0101311 | 12/2003 | Grohn | 398/154 | H04B 10/0771 |
| 2008/0260384 | 12/2007 | Tsuji et al. | N/A | N/A |
| 2016/0112143 | 12/2015 | Yu | 398/76 | H04B 10/6165 |
| 2017/0317750 | 12/2016 | Chaffee | N/A | H04B 10/1123 |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 2002341301 | 12/2001 | JP | N/A |
| 2005005897 | 12/2004 | JP | N/A |
| 2005070189 | 12/2004 | JP | N/A |

**OTHER PUBLICATIONS**

Office Action in related JP patent application 2023-546179, dated Jul. 23, 2024. cited by applicant
Search Report in related EP application 22745441.0, dated Mar. 25, 2024. cited by applicant
Deng KL et al: “All-optical clock extraction for self-clocked demultiplexing at 100 GB/s”, CLEO '97: Conference on Lasers and Electro-Optics. Baltimore, MD, May 18-23, 1997; [CLEO : Conference On Lasers and Electro-Optics], New York, IEEE, US, May 18, 1997 (May 18, 1997), pp. 412-413, XP032305568, DOI: 10.1109/CLEO.1997.603350 ISBN: 978-0-7803-4125-8. cited by applicant
Den K-L et al: “Unbalanced Toad for Optical Data and Clock Separation in Self-Clocked Transparent OTDM Networks”, IEEE Photonics Technology Letters, IEEE, USA, vol. 9, No. 6, Jun. 1, 1997 (Jun. 1, 1997), pp. 830-832, XP000692428, ISSN: 1041-1135, DOI:

10.1109/68.585005. cited by applicant
Yokoyama H et al: “Two-stage all-optical subharmonic clock recovery using modelocked semiconductor lasers”, Electronics Letters, the Institution of Engineering and Technology, GB, vol. 36, No. 18, Aug. 31, 2000 (Aug. 31, 2000), pp. 1577-1578, XP006015678, ISSN: 0013-5194, DOI: 10.1049/EL:20001124. cited by applicant

---

*Primary Examiner:* Wolf; Darren E

*Attorney, Agent or Firm:* Nathan & Associates Patent Agents Ltd.

---

## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 17/916,807 filed Oct. 4, 2022 (now allowed), which was a 371 application from international patent application PCT/IB2022/050063 filed Jan. 5, 2022, which claims the benefit of priority of U.S. Provisional patent application No. 63/142,684 filed Jan. 28, 2021, and which is incorporated herein by reference in its entirety.

FIELD
(1) Embodiments of various systems disclosed herein relate in general to signal transmission and more specifically to systems for transmission of high data rate optical signals.
BACKGROUND
(2) Coherent and non-coherent optical transceivers for high data rate transmissions such as 400 Gbps and beyond generally utilize digital signal processors (DSPs). The DSPs provide one or more of pre or post compensation of chromatic dispersion, other channel impairments, multiplexing and demultiplexing, implementation of high-level modulation formats and/or error correction to enable such high rates such as when using PAM4 (or other modulation).
(3) The required DSPs have high power consumption—typically 8W for a DSP in a system with a data rate of 400 Gbps, or higher power consumption for greater data rates, thus increasing the power requirements of high bitrate systems. Further, the multiplexing of multiple data channels and the other signal processing on the electrical signals introduces latency into the signal transmission. Still further, DSPs have a limited maximum throughput due to the constraints of electrical processing. It would be therefore be desirable to provide transmission systems featuring high data rates without the need for DSPs to thereby reduce the power consumption and latency and overcome the electrical bandwidth bottleneck.
SUMMARY
(4) Embodiments disclosed herein relate to systems, devices and methods that enable high data rate optical signals with reduced power consumption and reduced latency. The systems as described herein make use of all-optical transmitters and receivers such that no DSP is required, thereby reducing the power requirements and latency of the system for sending equivalent data rates. An all-optical transmitter receiver arrangement also reduces computational complexity (such as needed by a DSP). In some embodiments, the all-optical transmitters and receivers are provided as integrated circuits (ICs) on shared semiconductor substrates to further reduce power and size requirements.
(5) In order to provide an all-optical receiver, the clock (local oscillator) used in the transmitted modulated signal is needed at the receiver. A presently disclosed system uses a modulated pulsed laser for data transmission, utilizing the periodic pulses as a clock. The clock based on the pulsed laser is transmitted along with the data signal and recovered by the receiver to thereby serve as the required local oscillator. In some embodiments, orthogonal transverse modes are used, one carrying

the data and the other carrying the clock. In some embodiments, separate data and clock channels are transmitted to the receiver. To enable all-optical demodulation based on the received clock, the receiver advantageously includes a saturable absorber (SA) configured to transmit received data pulses when saturated by clock pulses to thereby recover the transmitted data signals.

(6) In some embodiments, optical time division multiplexing (OTDM) is provided for transmission of multiple data channels. Use of optical multiplexing reduces latency and complexity compared with electrical multiplexing such as in DSPs. In some embodiments, systems as disclosed herein may be duplicated or may be provided with integrated WDM systems for transmission of greater numbers of channels per fiber optic cable. In some embodiments, single mode fiber optic cable is used with unidirectional operation. In some embodiments, full duplex fiber optic cable is used with bidirectional operation. Further, dispersion management is also handled optically in the receiver to compensate for dispersion in transmission and for dispersion that may be caused by the saturable absorbers. As above, use of optical dispersion management reduces the computational complexity of the solution.

(7) In some embodiments, a system comprises: an optical transmitter that includes a pulsed laser for outputting a pulsed laser signal, wherein the optical transmitter is configured to modulate the pulsed laser signal based on a plurality of data signals to form a plurality of modulated optical data signals, to optically time division multiplex the plurality of modulated optical data signals to obtain a multiplexed optical data signal, and to transmit the multiplexed optical data signal together with an optical clock signal derived from the pulsed laser signal; and an optical receiver that includes a clock recovery branch and a data branch, wherein the clock recovery branch is configured to recover and phase-shift the optical clock signal into a plurality of phase-shifted recovered optical clock signals, wherein the data branch includes a plurality of SAs, and wherein each SA is in optical communication with one of the plurality of phase-shifted recovered optical clock signals and is configured to receive the multiplexed optical data signal and to transmit one of the modulated optical data signals received in the multiplexed optical data signal according to the one of the plurality of phase-shifted recovered optical clock signals.

(8) In some embodiments, the optical clock signal and the multiplexed optical data signal are transmitted by the optical transmitter in orthogonal transverse modes over a single mode fiber optic cable. In some embodiments, the optical clock signal and multiplexed optical data signal are each transmitted by the optical transmitter over a separate cable in a full-duplex fiber optic cable. In some embodiments, wherein a clock pulse from one of the plurality of phase-shifted recovered optical clock signals saturates a related SA to thereby cause the SA to release a pulse from one of the plurality of modulated optical data signals.

(9) In some embodiments, the clock recovery branch includes a splitter and a phase-shifter for extracting the optical clock signal into a plurality of phase-shifted extracted optical clock signals. In some embodiments, the optical receiver further includes one or more dispersion managers configured for dispersion correction of the plurality of modulated optical data signals and the plurality of phase-shifted extracted optical clock signals. In some embodiments, the optical receiver further comprises output components configured for extracting the plurality of data signals from each of the plurality of modulated optical data signals.

(10) In some embodiments, the pulsed laser is a selected from the group consisting of a bulk laser, a quantum well laser, a quantum dot laser, a semiconductor mode-locked laser and a mode-locked integrated external-cavity surface emitting laser. In some embodiments, the pulsed laser has a pulse repetition rate between 5 GHz to 100 GHz. In some embodiments, the optical receiver includes an optical amplifier (OA). In some embodiments, the OA is a quantum dot semiconductor optical amplifier.

(11) In some embodiments, the modulation is one of pulsed amplitude modulation (PAM) or quadrature amplitude modulation (QAM). In some embodiments, the plurality of data signals includes between 2-16 data signals per wavelength. In some embodiments, the plurality of data

signals are digital signals and/or analog signals.
(12) In some embodiments, a transmission system comprises a first transceiver and a second transceiver, wherein each of the first and second transceivers includes an optical transmitter, an optical receiver and a controller, wherein each optical transmitter is configured to modulate the pulsed laser signal based on a plurality of data signals to form a plurality of modulated optical data signals, to optically time division multiplex the plurality of modulated optical data signals to obtain a multiplexed optical data signal, and to transmit the multiplexed optical data signal together with an optical clock signal derived from the pulsed laser signal, and wherein each optical receiver includes a clock recovery branch and a data branch, wherein the clock recovery branch is configured to recover and phase-shift the optical clock signal into a plurality of phase-shifted recovered optical clock signals, wherein the data branch includes a plurality of SAs, and wherein each SA is in optical communication with one of the plurality of phase-shifted recovered optical clock signals and is configured to receive the multiplexed optical data signal and to transmit one of the modulated optical data signals received in the multiplexed optical data signal according to the one of the plurality of phase-shifted recovered optical clock signals.
(13) In some embodiments, each controller is configured for monitoring transceiver performance and for performing adjustment of the transmitter and receiver in order to optimize the transceiver performance. In some embodiments, the monitoring includes monitoring of one or more of bit error rate, data throughput, frame loss, or jitter. In some embodiments, each transceiver is formed on a common semiconductor substrate as an IC.
(14) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

---

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments disclosed herein are described, by way of example only, with reference to the following accompanying drawings, wherein:
(2) FIG. **1**A shows an optical system for data transmission over single mode fiber according to some embodiments;
(3) FIG. **1**B shows a transmitter for use with a single mode fiber according to some embodiments;
(4) FIG. **1**C shows illustrative signal amplitude graphs of the of the optical signals within the transmitter according to some embodiments;
(5) FIG. **1**D shows a receiver for use with a single mode fiber according to some embodiments;
(6) FIG. **1**E shows illustrative signal amplitude graphs of the optical signals within the receiver according to some embodiments;
(7) FIGS. **1**F and **1**G show multiple optical transmission systems according to some embodiments;
(8) FIG. **2**A shows an optical system for data transmission over full duplex fiber according to some embodiments;
(9) FIG. **2**B shows a transmitter for use with a full duplex fiber according to some embodiments;
(10) FIG. **2**C shows a coupling assembly for use with a full duplex fiber according to some embodiments;
(11) FIG. **2**D shows a receiver for use with a full duplex fiber according to some embodiments;
(12) FIGS. **2**E and **2**F show multiple optical transmission systems according to some embodiments.
DETAILED DESCRIPTION
(13) Aspects of embodiments relate to systems, devices and methods that enable high data rate optical signals with reduced power consumption, reduced latency and reduced computational

complexity. FIGS. **1**A-**1**G show an optical system for data transmission according to some embodiments. As shown in FIG. **1**A, an optical transmission system **100** includes transceivers **108-1** and **108-2**. Each transceiver **108** includes an optical transmitter **110**, an optical receiver **140**, and a controller **160**.

(14) Transmitter **110-1** is in optical communication with receiver **140-2** over a single mode fiber optic cable **170-1** for transmitting unidirectional data signals **102**, and transmitter **110-2** is in optical communication with receiver **140-1** over a single mode fiber optic cable **170-2** for transmitting unidirectional data signals **104**. In some embodiments, a single fiber optic cable **170** may be used for bidirectional optical communication such as by using different wavelengths for transmitting data signals **102** and **104**. Data signals, such as signals **102** and **104** are also referred to herein as “channels”. In some embodiments, each transceiver **108-1** and **108-2** is provided as an integrated circuit (IC) on a shared semiconductor substrate.

(15) Data signals **102**, **104** are electrical data signals. In some embodiments, data signals **102**, **104** are digital signals and the laser pulse repetition rate is equal or larger than the digital signal rate. In some embodiments, data signals **102**, **104** are analog electrical signals and the laser pulse repetition rate is equal or larger than the Nyquist frequency of the analog electrical signals. In some embodiments, data signals **102**, **104** are a combination of digital and analog signals. FIGS. **1**A-**1**G illustrate transmission of two channels in each transmission direction (data signals **102-1** and **102-2** and data signals **104-1** and **104-2**) for simplicity but it should be appreciated that more than two channels may be multiplexed and transmitted by system **100**. In some embodiments, between 2-16 data signals per wavelength are supported.

(16) A controller **160** is a computing device as defined herein. Controller **160-1** is in data communication with transmitter **110-1** and receiver **140-1**, and controller **160-2** is in data communication with transmitter **110-2** and receiver **140-2** for controlling each of transceivers **108-1** and **108-2** for the operation of system **100** as described further below.

(17) The components of each transmitter **110** are further described with reference to FIG. **1**B and include a pulsed laser **112**, a polarization splitter **114**, a 3 dB splitter **116**, drivers **118**, electro-optical (EO) modulators **120**, a tunable phase-shifter **122**, a 3 dB coupler **124**, a polarization coupler **126** and a pulse shaper **128**. In some embodiments, the components of transmitter **110** are formed on a common semiconductor as an IC.

(18) Pulsed laser **112** provides a source pulsed laser signal that may be modulated based on data signals **102** for optical transmission of data signals **102**. The pulsed laser signal is also used as a clock and is transmitted along with the modulated data to enable demodulation of the data signal. In some embodiments, pulsed laser **112** operates at a fixed wavelength here denoted λ.sub.1. Pulsed laser **112** emits pulses at a periodic repetition rate. In some embodiments, pulsed laser **112** may be any one of a bulk laser, quantum well laser, quantum dot laser, semiconductor mode-locked laser, high-power semiconductor laser or mode-locked integrated external-cavity surface emitting laser. In some embodiments, pulsed laser **112** may support a pulse repetition rate between 5 GHz to 100 GHz. In some embodiments, pulsed laser **112** may support a pulse repetition between 100 GHz-400 GHz.

(19) Polarization splitter **114** splits the pulsed laser signal into orthogonal transverse modes. In system **100**, one mode is used for data transmission and the other mode is used for clock transmission. The data is transmitted on a data branch **113** in the TE (transverse electric) mode and the clock is transmitted on a clock branch **111** in the TM (transverse magnetic) mode, or vice versa (data on TM and clock on TE). In some embodiments, polarization splitter **114** splits the pulsed laser signal strength unequally between the two modes. Non-limiting examples of split ratios include 90/10 and 80/20. In some embodiments, polarization splitter **114** splits the pulsed laser signal equally between the two modes. It is anticipated that both modes may suffer from the same distortions during transmission with no effect on the orthogonality.

(20) In data branch **113**, transmitter **112** provides for OTDM of multiple data signals **102** (or **104**).

Exemplarily, two data signals **102-1** and **102-2** as well as two drivers **118-1**, **118-2** and two EO modulators **120-1**, **120-2** are shown for simplicity, but it should be appreciated that multiple (more than two) data signals may be supported. 3 dB splitter **116** splits the pulsed transverse mode laser signal according to the number of data signals supported (here shown as two). In some embodiments, each data signal **102** is input into transmitter **112** via drivers **118**. Driver **118** may supply power to EO modulator **120** and may adjust the electrical signal level of the input data signal **102**.

(21) EO modulator **120** modulates the source pulsed laser signal based on input data signals **102**. In some embodiments, EO modulator may include a digital to analog converter (DAC) (not shown) for handling analog electrical inputs. In some embodiments, the pulse (clock) rate and thus the modulated channel rate is:

R/(number of channels)

(22) Various modulation schemes including but not limited to pulsed amplitude modulation (PAM) or quadrature amplitude modulation (QAM) can be applied. Tunable phase-shifter **122** shifts the phase of the second and further channels to obtain modulated (phase-shifted) channels (indicated as λ**1**.sub.S1-MOD, λ**1**.sub.S2-MOD, SHIFT) so that channel pulses will be interleaved when combined at 3 dB coupler **124**. Coupler **124** combines the modulated (phase-shifted) channels (indicated as λ**1**.sub.S1-MOD, λ**1**.sub.S2-MOD, SHIFT) to form an OTDM output signal (indicated as λ.sub.1 OTDM) of data branch **113** having a pulse rate of:

R×(number of channels)

(23) FIG. **1**C shows illustrative signal amplitude graphs of the modulated (phase-shifted) channels (indicted as λ**1**.sub.S1-MOD, λ**1**.sub.S2-MOD, SHIFT) and the combined OTDM output (indicated as λ.sub.1 OTDM) of data branch **113** as well as the clock signal (indicated as λ.sub.1 CLOCK).

(24) Polarization coupler **126** combines the transverse modes of the data and clock branches **111** and **113** to form orthogonal transverse signals carrying the OTDM output and the clock, indicated as λ**1**.sub.OTDM+CLOCK. In some embodiments, pulse shaper **128** shapes and equalizes the pulse heights and shapes of λ**1**.sub.OTDM+CLOCK to form the transmitter output **106**. In some embodiments, pulse shaper **128** may be coupled to a fiber optic cable **170**. In some embodiments, pulse shaper **128** may feed into a WDM multiplexer as describe further below. In some embodiments, pulse shaping may be implemented using filter banks. In some embodiments, pulse shaping may be implemented using nested unbalanced Mach-Zehnder interferometers (MZIs), ring resonators, or a combination of MZIs and ring resonators. Transmitter output **106** is transmitted on fiber optic cable **170** towards receiver **140**.

(25) In transmitter **110**, controller **160** may monitor and control all components and provide for automatic adjustment of adjustable components such as phase-shifters **122**, pulse shapers **128**, and drivers **118**.

(26) The components of each receiver **140** are further described with reference to FIG. **1**D and may include an optical amplifier (OA) **142**, a polarization splitter **144**, splitters **146**, polarization/phase controllers (PC) **148**, tunable phase-shifters **150**, dispersion managers **152**, saturable absorbers **154**, drivers **155**, circulators **156**, and output components **158**. In some embodiments, the components of receiver **140** may be formed on a common semiconductor substrate as an IC.

(27) OA **142** boosts the signal received from transmitter **110**. In some embodiments, OA **142** may be coupled to fiber optic cable **170**. In some embodiments, OA **142** may be connected to a WDM demultiplexer as described further below. In some embodiments, OA **142** may be a quantum dot semiconductor optical amplifier that has low power consumption and minimal cross talk between different wavelengths. In some embodiments, a driver **145** may supply power to OA **142**.

(28) The output of OA **142** (λ.sub.1 OTDM+CLOCK) is fed into polarization splitter **144**, which splits the received signal (λ.sub.1 OTDM+CLOCK) into orthogonal transverse modes. The data mode (TM or TE as defined in the transmitter) is transmitted on a data branch **141** and the clock mode (TM or TE as defined in the transmitter) is transmitted on a clock recovery branch **143**. In

some embodiments, polarization splitter **144** splits the received signal unequally between the two modes. Non-limiting examples of split ratios include 90/10 and 80/20. In some embodiments, polarization splitter **144** splits the received signal equally between the two modes.
(29) FIG. **1**E shows illustrative signal amplitude graphs of the combined OTDM signal (here indicated as λ.sub.1 OTDM) split within data branch **141**, the recovered clock and phase-shifted clocks (here indicated as λ.sub.1-CLOCK, λ.sub.1-CLOCK SHIFT) of clock branch **143** and the recovered modulated (phase-shifted) channels (here indicated as λ**1**.sub.S1-MOD, λ**1**.sub.S2-MOD, SHIFT).
(30) In data branch **141**, splitter **146-1** splits the OTDM according to the number of channels that were time division multiplexed by transmitter **110**. Thus, each split of data branch **141** carries λ.sub.1 OTDM. Similarly, in clock branch **143**, splitter **146-2** splits the clock signal λ.sub.1-CLOCK according to the number of channels that were time division multiplexed by transmitter **110**. As above, two OTDM channels and corresponding two clock channels are shown for simplicity, but it should be appreciated that multiple channels may be demultiplexed.
(31) In some embodiments, polarization/phase controllers (PC) **148** may be provided for matching the polarization and to correct phase mismatches. Exemplarily, multiple PCs **148-1** . . . **148-4** are provided, each on a separate signal or clock branch. Tunable phase-shifters **150-1** and **150-2** shift the phase of the clock channels so that clock pulses will be timed to coincide with the interleaved channels of the OTDM signal in data branch **141**. Each dispersion manager (DM) **152** may correct for dispersion in the optical channel that has affected the data and/or clock signals. Exemplarily, multiple DMs **152-1** . . . **152-4** are provided, each on a separate branch. DM **152** further may introduce pre-correction for dispersion introduced by a SA **154**. In some embodiments, a DM **152** may include chirped Bragg gratings controlled by a temperature controller **153**. In some embodiments, a driver **155** may supply power to a respective SA **154**.
(32) Each output of clock branch **143** (here indicated as λ.sub.1 CLOCK and λ.sub.1 CLOCK SHIFT) is fed into a corresponding circulator **156** in the data branch **141**. Exemplarily, two circulators **156-1**, **156-2**, two SAs **154-1**, **154-2**, and two drivers **155-1**, **155-2** are shown corresponding to two multiplexed data channels. In some embodiments, circulators **156** may be non-reciprocating, one directional, three-port circulators. In FIG. **1**D, circulators **156** are shown as having a clockwise direction, but it should be appreciated that the direction employed will be functional. In use, the clock input is circulated to an SA **154**. In some embodiments, each SA **154** may be any one of a quantum well (QW) semiconductor SA, a quantum dot SA, or a bulk semiconductor SA.
(33) A clock pulse will saturate SA **154**, forcing SA **154** to be transparent to thus “release” or “transmit” the current OTDM interleaved channel pulse through SA **154** corresponding to the pulse period. Thus, λ.sub.1 CLOCK and λ.sub.1 CLOCK SHIFT will each respectively saturate SA **154-1** and SA **154-2** at different time periods, each time period corresponding to a separate OTDM channel, to thereby release the current OTDM channel pulse through SA **154**. When the clock signal is between pulses (low clock signal), SA **154** is not saturated, thus blocking other OTDM channels. In some embodiments, each SA **154-1** or **154-2** provides an electrical output of each recovered clock **159-1** or **159-2**.
(34) The released OTDM pulse will enter circulator **156** and will be circulated to output components (OC) **158**. Each OC **158** includes those components necessary for converting the released optical channel back into an electrical data signal **102**. Exemplarily, two OCs **158-1**, **158-2** are shown, corresponding to two multiplexed data channels. OC **158** may include one or more of a photodiode (PD) or a PD array for optical-to-electrical conversion, a transimpedance amplifier (TIA), a DAC, and/or a limiting amplifier (LA).
(35) In receiver **140**, controller **160** may monitor and control all components and provide for automatic adjustment of adjustable components such as phase-shifters **150**, OA **142**, drivers **145**, **155**, and DMs **152**. In some embodiments, controller **160** monitors transceiver **108** performance in

terms of one or more of bit error rate, data throughput, frame loss, jitter, and other parameters, and then performs adjustment of adjustable components (in both of transmitter **110** and receiver **140**) in order to optimize the transceiver **108** performance.

(36) As shown in FIG. **1**F, in some embodiments, multiple systems **100-1**, **100-2**, to **100**-X (where X=N/2) may be provided in parallel, each system **100** operating over a separate pair of fiber optic cables **170-1** and **170-2**. As further shown, each of systems **100** may provide OTDM of up to four channels in each direction (**102**, **104**). In some embodiments, more than four channels may be provided. Each of systems **100-1**, **100-2**, **100**-X may operate using a pulsed laser **112** with the same wavelength or different wavelengths or a combination of wavelengths.

(37) As shown in FIG. **1**G, in some embodiments, multiple systems **100-1** to **100**-X may be provided in parallel along with a wavelength division multiplexing (WDM) multiplexers **172-1** and **174-2** and demultiplexers **172-2** and **174-1**, each operating over a respective single fiber optic cable **170-1** and **170-2**. For operation using WDM, each of systems **100-1**, **100-2** . . . **100**-N uses a respective pulsed laser **112** with a different wavelength. As further shown in FIG. **1**G, each of systems **100** may provide OTDM of up to four channels in each direction (**102**, **104**). In some embodiments, more than four channels may be provided.

(38) FIGS. **2**A-**2**F show an optical system for bidirectional data transmission over a full-duplex fiber according to some embodiments. As shown in FIG. **2**A, an optical transmission system **200** includes transceivers **208-1** and **208-2**. Each transceiver **208** includes an optical transmitter **210**, an optical receiver **240**, a coupling assembly **209**, and a controller **260**.

(39) Transmitter **210-1** is in optical communication with receiver **240-2** over a full duplex fiber optic cable **270** for transmitting data signals **102** using a first wavelength (λ.sub.1), and transmitter **210-2** is in optical communication with receiver **240-1** over full duplex fiber optic cable **270** for transmitting data signals **104** using a second wavelength (λ.sub.2). In some embodiments, each of transceivers **208** is provided as an IC on a shared semiconductor substrate.

(40) Data signals **102**, **104** are electrical data signals. In some embodiments, data signals **102**, **104** are digital signals and the laser pulse repetition rate is equal or larger than the digital signal rate. In some embodiments, data signals **102**, **104** are analog electrical signals and the laser pulse repetition rate is equal or larger to the Nyquist frequency of the analog electrical signals. In some embodiments, data signals **102**, **104** are a combination of digital and analog signals. FIGS. **2**A-**2**F illustrate transmission of two channels in each transmission direction (data signals **102-1** and **102-2** and data signals **104-1** and **104-2**) for simplicity but it should be appreciated that more than two channels may be transmitted by system **200**. In some embodiments, between 2-16 data signals per wavelength are supported.

(41) Each controller **260** is a computing device as defined herein. Controller **260-1** is in data communication with transmitter **210-1** and receiver **240-1** and controller **260-2** is in data communication with transmitter **210-2** and receiver **240-2** for controlling each of transceivers **208-1** and **208-2** for the operation of system **200** as described further below. In some embodiments, each of coupling assembly **209-1**, **209-2** is in data communication with its respective controller **260-1**, **260-2**.

(42) The components of each transmitter **210** are further described with reference to FIG. **2**B and include a pulsed laser **112**, a splitter **214**, a 3 dB splitter **116**, drivers **118**, EO modulators **120**, a tunable phase-shifter **122**, a 3 dB coupler **124**, and pulse shapers **228**. In some embodiments, the components of transmitter **210** are formed on a common semiconductor as an IC. Like numbered components of transmitter **110** are the same components and provide the same functionality in transmitter **210** and are therefore not described again.

(43) Splitter **214** splits the pulsed laser signal into a data branch **213** for preparing an OTDM signal for transmission and into a clock branch **211** for clock transmission. In some embodiments, splitter **214** splits the pulsed laser signal strength unequally between the two branches. Non-limiting examples of split ratios include 90/10 and 80/20. In some embodiments, splitter **214** splits the

pulsed laser signal equally between the two branches.
(44) In data branch **213**, transmitter **212** provides for OTDM of multiple data signals **102** (or **104**). Two data signals **102-1** and **102-2** are shown for simplicity but it should be appreciated that multiple data signals may be supported. Data branch **213** functionality is the same as for system **100** and forms the OTDM output (indicated as MI OTDM) of data branch **213** having a pulse rate of R×(number of channels).
(45) In some embodiments, pulse shaper **228-1** shapes and equalizes the pulse heights and shapes of λ.sub.1 OTDM. In some embodiments, pulse shaper **228-2** shapes and equalizes the pulse heights of λ.sub.1 CLOCK. In some embodiments, pulse shaping may be implemented using filter banks. In some embodiments, pulse shaping may be implemented using nested unbalanced MZIs or ring resonators or a combination of MZIs and ring resonators.
(46) As shown in FIG. **2**C, the output of transmitter **210** feeds into coupling assembly **209**. Each coupling assembly **209** includes circulators **230** and couplers **232**. Pulse shaper **228-1** feeds λ.sub.1 OTDM into optical circulator **230-1** and pulse shaper **228-2** feeds λ.sub.1 CLOCK into optical circulator **230-2**. In some embodiments, circulators **230** may be a non-reciprocating, one directional, three-port circulators. In FIG. **2**C circulators **230** are shown as having a clockwise direction, but It should be appreciated that the direction employed will be functional.
(47) One of the circulator **230** ports is connected to couplers **232**. Coupler **232-1** couples the outgoing (Tx) λ.sub.1 OTDM to a first of the fibers in cable **270** and coupler **232-2** couples outgoing λ.sub.1 CLOCK to a second of the fibers in fiber optic cable **270**. The incoming (Rx) clock and OTDM signals λ.sub.2 OTDM and λ.sub.2 CLOCK from cable **270**, exit couplers **232** and are directed by circulators **230** to exit ports of circulators **230** that are coupled to a “same-side” receiver **240** input.
(48) In transmitter **210**, controller **160** may monitor and control all components and provide for automatic adjustment of adjustable components such as phase-shifters **122**, pulse shapers **228**, and drivers **118**.
(49) The components of each receiver **140** are further described with reference to FIG. **2**D and may include optical amplifiers (OA) **242**, splitters **146**, polarization/phase controllers (PC) **148**, tunable phase-shifters **150**, dispersion managers **152**, saturable absorbers **154**, drivers **155**, circulators **156**, and output components **158**. In some embodiments, the components of receiver **240** may be formed on a common semiconductor as an IC. Like numbered components of receiver **140** are the same components and provide the same functionality in receiver **240** and are therefore not described again.
(50) OA **242-1** boosts the OTDM signal received from transmitter **210**. OA **242-2** boosts the clock signal received from transmitter **210**. Both of OAs **242** may be coupled to fiber optic cable **270** via coupling assembly **209**. In some embodiments, OA **242** may be connected to a WDM demultiplexer as described further below. In some embodiments, OA **242** may be a quantum dot semiconductor optical amplifier that has low power consumption and minimal cross talk between different wavelengths. In some embodiments, drivers **245** may supply power to OAs **242**.
(51) The output of OA **242-1** is fed into a data branch **241** and the output of OA **242-2** is transmitted on a clock branch **243**. The functionality and operation of data branch **241** are the same as data branch **141** as described above, and the functionality and operation of clock branch **243** are the same as clock branch **143** as described above.
(52) In receiver **240**, controller **260** may monitor and control all components and provide for automatic adjustment of adjustable components such as phase-shifters **150**, OAs **242**, drivers **155**, and DMs **152**. In some embodiments, controller **260** monitors transceiver **208** performance in terms of one or more of bit error rate, data throughput, frame loss, jitter, and other parameters, and then performs adjustment of adjustable components (in both of transmitter **210** and receiver **240**) in order to optimize the transceiver **208** performance.
(53) As shown in FIG. **2**E, in some embodiments, multiple systems **200-1**, **200-2**, to **200**-X (where

X=N/2) may be provided in parallel each operating over a separate full duplex fiber optic cable **270-1**, **270-2** . . . **270**-X. As further shown, each of systems **200** may provide OTDM of up to four channels in each direction (**102**, **104**). In some embodiments, more than four channels may be provided. Each of systems **200-1**, **200-2** . . . **200**-X may operate using a pulsed laser **112** with the same wavelength or different wavelengths or a combination of wavelengths.
(54) As shown in FIG. **2**F, in some embodiments, multiple systems **200** may be provided in parallel along with a dual WDM multiplexers **272-1** and **272-2** and demultiplexers **274-1** and **274-2** operating over a full duplex fiber optic cable **270**. For simplicity only one direction of transmission is shown in FIG. **2**F. OTDM signals are transmitted using WDM pair **272-1** and **274-1** and clock signals are transmitted using WDM pair **272-2** and **274-2**. In some embodiments, one OA **242** is positioned before demultiplexer **274-1** and one OA **242** is positioned before demultiplexer **274-2**. For operation using WDM, each of systems **200-1**, **200-2** . . . **200**-X uses a pulsed laser **112** with a different wavelength. As further shown, each of systems **200** may provide OTDM of up to four channels. In some embodiments, more than four channels may be provided.
(55) In the claims or specification of the present application, unless otherwise stated, adjectives such as “substantially” and “about” modifying a condition or relationship characteristic of a feature or features of an embodiment of the invention, are understood to mean that the condition or characteristic is defined to within tolerances that are acceptable for operation of the embodiment for an application for which it is intended.
(56) Implementation of the method and system of the present disclosure may involve performing or completing certain selected tasks or steps manually, automatically, or a combination thereof. Moreover, according to actual instrumentation and equipment of preferred embodiments of the method and system of the present disclosure, several selected steps may be implemented by hardware (HW) or by software (SW) on any operating system of any firmware, or by a combination thereof. For example, as hardware, selected steps of the disclosure could be implemented as a chip or a circuit. As software or algorithm, selected steps of the disclosure could be implemented as a plurality of software instructions being executed by a computer using any suitable operating system. In any case, selected steps of the method and system of the disclosure could be described as being performed by a data processor, such as a computing platform for executing a plurality of instructions.
(57) Although the present disclosure is described with regard to a computing device, or a computer, it should be noted that optionally any device featuring a data processor and the ability to execute one or more instructions may be described as a computing device, including but not limited to any type of personal computer (PC), a server, a distributed server, a master control unit, a virtual server, a cloud computing platform, a cellular telephone, an IP telephone, a smartphone, a smart watch or a PDA (personal digital assistant). Any two or more of such devices in communication with each other may optionally form a “network” or a “computer network”.
(58) It should be understood that where the claims or specification refer to “a” or “an” element, such reference is not to be construed as there being only one of that element. In the description and claims of the present application, each of the verbs, “comprise” “include” and “have”, and conjugates thereof, are used to indicate that the object or objects of the verb are not necessarily a complete listing of components, elements or parts of the subject or subjects of the verb.
(59) While this disclosure describes a limited number of embodiments, it will be appreciated that many variations, modifications and other applications of such embodiments may be made. The disclosure is to be understood as not limited by the specific embodiments described herein, but only by the scope of the appended claims.

## Claims

1. A method, comprising: providing an optical transmitter having a pulsed laser, an optical receiver having a data branch with a plurality of saturable absorbers (SA) and a clock branch, and a polarization splitter; by the pulsed laser, outputting a pulsed laser signal; modulating the pulsed laser signal based on a plurality of data signals to form a plurality of modulated optical data signals; optically time division multiplexing the plurality of modulated optical data signals to obtain a multiplexed optical data signal; by the polarization splitter, splitting the pulsed laser signal into the data branch and into the clock branch for transmitting the multiplexed optical data signal through the data branch and for transmitting an optical clock signal separately through the clock branch, wherein the clock branch is configured to recover and phase-shift the optical clock signal into a plurality of phase-shifted recovered optical clock signals, and wherein each of the SAs is in optical communication with one of the plurality of phase-shifted recovered optical clock signals; and configuring each of the SAs to receive the multiplexed optical data signal and to demultiplex one of the modulated optical data signals according to a respective optical clock signal from the plurality of phase-shifted recovered optical clock signals.
2. The method of claim 1, wherein the transmitting of the multiplexed optical data signal and of the optical clock signal includes transmitting the multiplexed optical data signal and the optical clock signal in orthogonal transverse modes over a single mode fiber optic cable.
3. The method of claim 1, wherein the transmitting of the multiplexed optical data signal and of the optical clock signal includes transmitting the multiplexed optical data signal and the optical clock signal over a separate cable in a full-duplex fiber optic cable.
4. The method of claim 1, further comprising saturating a respective one of the SAs to thereby cause the respective one of the SAs to release a pulse from one of the plurality of modulated optical data signals.
5. The method of claim 1, further comprising extracting the optical clock signal into the plurality of phase-shifted recovered optical clock signals.
6. The method of claim 1, further comprising correcting dispersion related to each of the plurality of modulated optical data signals and the plurality of phase-shifted recovered optical clock signals.
7. The method of claim 1, further comprising extracting the plurality of data signals from each of the plurality of modulated optical data signals.
8. The method of claim 1, wherein the modulating of the pulsed laser signal includes modulating using pulsed amplitude modulation (PAM) or quadrature amplitude modulation (QAM).
9. The method of claim 1, wherein the pulsed laser is selected from the group consisting of a bulk laser, a quantum well laser, a quantum dot laser, a semiconductor mode-locked laser and a mode-locked integrated external-cavity surface emitting laser.
10. The method of claim 1, wherein the pulsed laser has a pulse repetition rate between 5 GHz to 100 GHz.
11. The method of claim 1, wherein the optical receiver includes an optical amplifier (OA).
12. The method of claim 11, wherein the OA is a quantum dot semiconductor optical amplifier.
13. The method of claim 1, wherein the plurality of data signals includes between 2-16 data channels.
14. The method of claim 1, wherein the plurality of data signals includes digital signals and/or analog signals.